

$$I_{D3,4} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{3,4} (V_{ov3,4})^2$$

$$50 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times 50 \times (V_{ov3,4})^2$$

$$V_{ov3,4} = 0.14 \text{ V}$$

$$\therefore |V_{GS3,4}| = 0.14 + 0.5 = 0.64 \text{ V}$$

$$\therefore V_{b4} = V_{DD} - |V_{GS3,4}| = 1.8 - 0.64 = 1.16 \text{ V}$$

$$I_{D10} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{10} (V_{ov10})^2$$

$$40 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times 50 \times (V_{ov10})^2$$

$$V_{ov10} = 0.09 \text{ V}$$

$$V_{GS10} = 0.09 \text{ V} + 0.4 = 0.49 \text{ V}$$

$$V_p = V_{b1} - V_{GS10} = 0.6 - 0.49 = 0.11 \text{ V}$$

$$V_p = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{out1} + V_{out2} - 2V_{thn})} \times 2I_D$$

$$0.11 = \frac{1}{200 \times 10^{-6} \times \left(\frac{W}{L} \right)_{11,12} (1.8 - 2 \times 0.4)} \times 2 \times 40 \times 10^{-6}$$

$$\left(\frac{W}{L} \right)_{11,12} = 3.63$$

$$V_{out, min} = V_p + V_{ovg} + V_{ov7}$$

$$= 0.11 + 0.09 + 0.09 = 0.29 \text{ V}$$

$$R_{out, down} = g_{m7} r_{o7} r_{o9}$$

$$g_{m7} = \frac{2I_D}{V_{ov7}} = \frac{2 \times 40 \times 10^{-6}}{0.09} = 0.89 \text{ mS}$$

$$r_{o7} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 40 \times 10^{-6}} = 250 \text{ k}\Omega$$

$$r_{o9} = 250 \text{ k}\Omega$$

$$R_{out, down} = 0.89 \text{ mS} \times 250 \text{ k}\Omega \times 250 \text{ k}\Omega = 55.63 \text{ M}\Omega$$

$$I_{D5} = 0.8 I_{D3} = 40 \mu\text{A}$$

$$\therefore I_{D3} = 50 \mu\text{A}$$

$$\frac{V_{out1} + V_{out2}}{2} = 0.5 V_{DD}$$

$$\therefore V_{out1} + V_{out2} = V_{DD} = 1.8$$

PART B

$$I_{D3,4} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{3,4} (V_{OV3,4})^2$$

$$80 \times 10^{-6} = \frac{1}{2} \times 80 \times 10^{-6} \times 40 \times (V_{OV3,4})^2$$

$$V_{OV3,4} = 0.22 \text{ V}$$

$$\therefore |V_{GS3,4}| = 0.22 + 0.5 = 0.72$$

$$\therefore V_{D4} = V_{DD} - |V_{GS3,4}| = 1.8 - 0.72 = 1.08 \text{ V}$$

$$I_{D10} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{10} (V_{OV10})^2$$

$$50 \times 10^{-6} = \frac{1}{2} \times 150 \times 10^{-6} \times 40 \times (V_{OV10})^2$$

$$V_{OV10} = 0.13 \text{ V}$$

$$V_{GS10} = 0.13 + 0.4 = 0.53$$

$$V_P = V_{b1} - V_{GS10} = 0.6 - 0.53 = 0.07 \text{ V}$$

$$V_P = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{out1} + V_{out2} - 2V_{thn})} \times 2I_D$$

$$0.07 = \frac{1}{150 \times 10^{-6} \times \left(\frac{W}{L} \right)_{11,12} (2.16 - 2 \times 0.4)} \times 2 \times 50 \times 10^{-6}$$

$$\left(\frac{W}{L} \right)_{11,12} = 7.00$$

$$V_{out, min} = V_P + V_{OV3} + V_{OV7}$$

$$= 0.07 + 0.13 + 0.13 = 0.33 \text{ V}$$

$$R_{out, down} = g_{m7} r_{o7} r_{o3}$$

$$g_{m7} = \frac{2I_D}{V_{OV}} = \frac{2 \times 50 \times 10^{-6}}{0.13} = 0.77 \text{ mS}$$

$$r_{o7} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 50 \times 10^{-6}} = 200 \text{ k}\Omega$$

$$r_{o3} = 200 \text{ k}\Omega$$

$$R_{out, down} = 0.77 \text{ mS} \times 200 \text{ k}\Omega \times 200 \text{ k}\Omega = 30.8 \text{ M}\Omega$$

$$I_{D5} = 0.625 I_{D3} = 50 \mu\text{A}$$

$$\therefore I_{D3} = 80 \mu\text{A}$$

$$\frac{V_{out1} + V_{out2}}{2} = 0.6 V_{DD}$$

$$\therefore V_{out1} + V_{out2} = 1.2 V_{DD} = 2.16$$

PART C

$$I_{D3,4} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{3,4} (V_{ov3,4})^2$$

$$120 \times 10^{-6} = \frac{1}{2} \times 60 \times 10^{-6} \times 60 \times (V_{ov3,4})^2$$

$$V_{ov3,4} = 0.26$$

$$\therefore |V_{GS3,4}| = 0.26 + 0.45 = 0.71$$

$$\therefore V_{b4} = V_{DD} - |V_{GS3,4}| = 1.8 - 0.71 = 1.09 \text{ V}$$

$$I_{D10} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{10} (V_{ov10})^2$$

$$60 \times 10^{-6} = \frac{1}{2} \times 120 \times 10^{-6} \times 60 \times (V_{ov10})^2$$

$$V_{ov10} = 0.13 \text{ V}$$

$$V_{GS10} = 0.13 + 0.3 = 0.43 \text{ V}$$

$$V_p = V_{b1} - V_{GS10} = 0.55 - 0.43 = 0.12 \text{ V}$$

$$V_p = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{out1} + V_{out2} - 2V_{thn})} \times 2I_D$$

$$0.12 = \frac{1}{120 \times 10^{-6} \times \left(\frac{W}{L} \right)_{11,12} (1.44 - 2 \times 0.3)} \times 2 \times 60 \times 10^{-6}$$

$$\left(\frac{W}{L} \right)_{11,12} = 9.92$$

$$V_{out, min} = V_p + V_{ov9} + V_{ov7}$$

$$= 0.12 + 0.13 + 0.13 = 0.38 \text{ V}$$

$$R_{out, down} = g_{m7} r_{o7} r_{o9}$$

$$g_{m7} = \frac{2I_D}{V_{ov}} = \frac{2 \times 60 \times 10^{-6}}{0.13} = 0.92 \text{ mS}$$

$$r_{o7} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 60 \times 10^{-6}} = 166.67 \text{ k}\Omega$$

$$r_{o9} =$$

$$R_{out, down} = 0.92 \text{ mS} \times 166.67 \text{ k}\Omega \times 166.67 \text{ k}\Omega = \underline{\underline{25.66 \text{ M}\Omega}}$$

$$I_{D5} = 0.5 I_{D3} = 60 \text{ }\mu\text{A}$$

$$\therefore I_{D3} = 120 \text{ }\mu\text{A}$$

$$\frac{V_{out1} + V_{out2}}{2} = 0.4 V_{DD}$$

$$\therefore V_{out1} + V_{out2} = 0.8 V_{DD} = 1.44 \text{ V}$$